NeuroGate: Achieving Transformer-Parity with Biologically-Grounded Dendritic Dynamics

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Abstract

The artificial neuron has remained architecturally unchanged since Rosenblatt's perceptron [Rosenblatt, 1958]: it computes a static function of its current input, possessing no intrinsic temporal memory. Biological cortical neurons, by contrast, perform multi-timescale integration across anatomically distinct dendritic compartments. Inspired by experimental neuroscience—particularly vectorized instructive signals in cortical apical dendrites [Francioni et al., 2026] and first-spike temporal coding [Thorpe et al., 2001]—we introduce **NeuroGate**, a principled redesign of the fundamental artificial neuron unit around four biologically-grounded mechanisms: (1) dual-timescale exponential memory traces for fast (Na^+, α_{fast}) and slow (Ca^{2+}, α_{slow}) dendritic dynamics; (2) input projection before temporal integration; (3) a context-dependent write gate (NMDA-LTP selective encoding); and (4) hard-wired shortcuts to anchor positions (first-spike coding). All parameters are learned end-to-end with no manually tuned constants.

We evaluate NeuroGate through an ablation study (V1–V6), three benchmark tasks, and an architecture-level test. On the Delayed Sign-XOR task, NeuroGate achieves 99.2% accuracy converging in 2 epochs—four times faster than LSTM—using 40% fewer parameters, while the perceptron is architecturally incapable of exceeding 52.3% chance. On Sequential MNIST, NeuroGate achieves 2.1× higher accuracy than LSTM using 46% fewer parameters at matched training budget. On the multi-lag regression task, LSTM outperforms NeuroGate, correctly characterising NeuroGate's domain as temporal *pattern detection* rather than temporal *value storage*.

For the architecture test, we replace Multi-Head Attention entirely with stacked NeuroGate layers (**NeuroGate-Linear**) on TinyShakespeare character-level language modelling ($T = 32$). At matched parameter count (∼807K vs ∼804K), NeuroGate-Linear performs within 2% of the Transformer (PPL 6.21 vs 6.00), demonstrating that biological temporal inductive bias can *substitute* for attention with comparable quality. With 43% more parameters, NeuroGate-Linear outperforms a Transformer by 14.3% in validation loss (PPL 4.23 vs 5.39), confirmed in 5 seeds (bootstrap 95% CI: [−15.3%, −13.5%]). A parallel `conv1d` implementation runs 5.3× faster with statistically identical results.

In a preliminary extension (**Dopa-SSM**), we add dopaminergic-inspired neuromodulatory input-dependent dynamics and expand from two to four learned dendritic timescales. Dopa-SSM outperforms both the Transformer (−2.6%, PPL 5.92 vs 6.21) and a simplified Mamba baseline (−0.3%) on TinyShakespeare ($T = 64$), while running 3.6× faster than Mamba via fully parallel `conv1d` computation.

Post-training inspection confirms $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ in every layer of every experiment without any constraint, reproducing the biological timescale hierarchy through gradient descent alone. These results demonstrate that a biologically-grounded temporal neuron can serve as a viable alternative to attention at the scales tested, and that the perceptron's temporal blindness is a fundamental architectural limitation—not merely a training challenge.

Keywords: biologically-inspired neural networks, temporal sequence modelling, dendritic computation, attention-free architecture, first-spike coding, neuromodulation, NeuroAI

1 Introduction

Standard artificial neurons compute $y = \sigma(\mathbf{w}^\top \mathbf{x}_t + b)$, operating only on the current input $\mathbf{x}_t$. Despite decades of architectural innovation—residual connections [He et al., 2016], attention [Vaswani et al., 2017], normalisation [Ba et al., 2016]—the fundamental unit itself has not been reconsidered. Temporal complexity is built *around* the perceptron at the network level, not *into* the neuron.

Biological cortical pyramidal neurons operate very differently. They have two anatomically and functionally distinct integration zones: proximal (basal) dendrites integrate inputs arriving at the perisomatic region on a fast timescale ($\sim$5 ms), while distal apical dendrites integrate feedback from higher cortical areas on a slow timescale ($\sim$100 ms) [Larkum, 2013, Losonczy and Magee, 2006]. The apical compartment acts as a teaching signal: Francioni et al. [2026] showed experimentally that it carries per-neuron vectorized instructive signals during learning, and that optogenetically blocking these signals abolishes learning entirely.

This biological architecture solves at the single-neuron level the same temporal integration problem that attention mechanisms address at the network level. Our central hypothesis is:

> *Multi-Head Attention in Transformers exists partly to compensate for the perceptron's temporal blindness. Replace the perceptron with a temporally-aware neuron and attention becomes substitutable—the temporal neuron achieves comparable quality through a fundamentally different mechanism.*

We test this by building NeuroGate through a systematic ablation study, then replacing attention entirely in a language model.

1.1 Contributions

(1) **Systematic neuron ablation (V1–V6).** We design NeuroGate through six iterations, each adding exactly one biological mechanism, with full empirical attribution of each gain (Table 1).

(2) **Gradient barrier identification.** We show empirically that EMA-based neurons suffer gradient attenuation of $(1-\alpha)\alpha^{T-1-t}$ to early timesteps—only 2.5% at $T = 8$, $\alpha = 0.7$—blocking learning from early history regardless of training duration.

(3) **Multi-anchor shortcuts.** We introduce N direct gradient bypasses at uniformly spaced positions, inspired by first-spike coding [Thorpe et al., 2001]. Adding one anchor at $t = 1$ raises XOR accuracy from 73% to 99.2%.

(4) **Attention-free language modelling.** NeuroGate-Linear replaces Multi-Head Attention entirely. At matched parameter count ($\sim$807K vs $\sim$804K), NeuroGate-Linear performs within 2% of the Transformer (PPL 6.21 vs 6.00), demonstrating that temporal inductive bias can substitute for attention at comparable quality.

(5) **Parallel GPU implementation.** We prove that sequential EMA $h_t = \alpha h_{t-1} + b_t$ equals causal convolution $h_t = \sum_{k \le t} \alpha^{t-k} b_k$, enabling a `conv1d`-based implementation that is 5.3× faster with statistically identical results.

(6) **Unsupervised biological validation.** Gradient descent independently recovers $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ in every layer of every experiment, matching the Ca^{2+}/Na^{+} timescale hierarchy reported in the neuroscience literature.

2 Related Work

Gated recurrent models. LSTM [Hochreiter and Schmidhuber, 1997] and GRU [Cho et al., 2014] introduced gated state cells but were not biologically motivated. Recent linear recurrent models—Mamba [Gu and Dao, 2023], RWKV [Peng et al., 2023], RetNet [Sun et al., 2023]—pursue efficiency through state-space formulations but retain monolithic hidden states. Linear Recurrent Units (LRU; Orvieto et al. 2023) and xLSTM [Beck et al., 2024] extend gated recurrence with diagonal state-space parameterisations and exponential gating respectively. Structured state space models (S4, HiPPO; Gu et al. 2020, 2022) learn parametric timescales via polynomial projections. Liquid Neural Networks [Hasani et al., 2021] draw on ODE-based neural dynamics inspired by the *C. elegans* connectome. NeuroGate differs from all of these in two respects: every architectural component maps to a specific experimental neuroscience finding in cortical pyramidal neurons, and the anchor shortcuts (Section 3.2) provide direct gradient access to arbitrary historical positions—a mechanism with no equivalent in state space models.

Dendritic computation. Larkum [2013] established the basal/apical functional distinction in cortical pyramidal neurons. Losonczy and Magee [2006] characterised fast Na^+ dendritic spikes and slow Ca^{2+} plateau potentials as two distinct integration mechanisms—the direct inspiration for NeuroGate's dual-timescale design. Francioni et al. [2026] demonstrated experimentally that apical dendrites carry vectorized per-neuron instructive signals, motivating NeuroGate's separate slow-context EMA pathway.

First-spike coding. Thorpe et al. [2001] showed that object recognition in the primate visual system relies on the relative timing of first spikes rather than firing rates. Larkum et al. [1999] demonstrated that the first action potential can trigger a Ca^{2+} burst that potentiates backpropagating signals—providing the biological mechanism underlying NeuroGate's anchor shortcut.

Attention-free architectures. Linear attention [Katharopoulos et al., 2020], FNet [Lee-Thorp et al., 2022], and gated convolutional networks [Dauphin et al., 2017] have explored replacing attention with linear operations, with mixed results. None are grounded in dendritic neuroscience. NeuroGate-Linear is, to our knowledge, the first attention-free architecture with explicit biological grounding that achieves near-parity with standard Transformers at matched parameter count.

NeuroAI. Zador et al. [2023] argued for integrating biological structural inductive biases into AI systems. Our work implements this programme at the most fundamental level: redesigning the basic computational unit.

3 The NeuroGate Architecture

3.1 Ablation Study: V1–V6

Table 1 shows the effect of adding each biological mechanism. Each version adds exactly one component; all other hyperparameters remain identical. Accuracy is reported for the Delayed Sign-XOR task (Section 4.1).

3.2 Architecture

Let $\mathbf{x} \in \mathbb{R}^{T \times n_{\text{in}}}$ be an input sequence. NeuroGate computes its output through the following sequence of operations.

Table 1: Ablation study: V1–V6, each version adds one biological mechanism. Delayed Sign-XOR accuracy ($T = 8$, $n_{\text{hid}} = 32$, 80 epochs, seed 42). For multi-seed validation of the final NeuroGate model, see Table 7. STDP (V5) does not improve XOR, indicating that structural gradient access matters more than correlation statistics on this task.

Version	Mechanism	Biological reference	XOR acc.
V1 Perceptron	None (baseline)	—	52.3%
V2 Decay gate	Exponential decay context	RC membrane	53.8%
V3 Dual EMA	Two timescales (fast + slow)	Losonczy and Magee [2006]	64.2%
V4 Projected EMA	Input projection before EMA	Larkum [2013]	73.0%
V5 STDP	Temporal correlation gate	Bi and Poo [1998]	71.3%
V6 NeuroGate	**First-spike anchor + write gate**	Thorpe et al. [2001]	**99.2%**
LSTM ($h = 32$)	Standard LSTM	Hochreiter and Schmidhuber [1997]	98.2%

Step 1: Input projection.

$$\tilde{\mathbf{x}}_t = \tanh(\mathbf{W}_{\text{proj}}\,\mathbf{x}_t + \mathbf{b}_{\text{proj}}) \tag{1}$$

Biological analogy: synaptic currents transform pre-synaptic signals before entering the dendritic compartment [Larkum, 2013].

Step 2: Context signal. A lightweight causal EMA serves as a context signal for the write gate:

$$\mathbf{c}_t = \sum_{k=0}^{t} \alpha_c^{t-k}\,\mathbf{x}_k, \qquad \alpha_c = e^{-\text{softplus}(\tau_{c,\text{raw}})} \tag{2}$$

Step 3: Context-dependent write gate.

$$\mathbf{w}_t = \sigma(\mathbf{W}_{\text{wx}}\,\mathbf{x}_t + \mathbf{W}_{\text{wc}}\,\mathbf{c}_t) \tag{3}$$

Biological analogy: NMDA receptors act as coincidence detectors, gating synaptic potentiation when pre-synaptic input coincides with dendritic depolarisation [Malenka and Nicoll, 1993]. Note: $\mathbf{c}_t$ (Eq. 2) modulates the gate using a *separate* context pathway, not self-feedback—consistent with apical/basal anatomical separation [Larkum, 2013].

Step 4: Write-gated dual-timescale EMA. For each timescale $\tau \in \{\tau_{\text{fast}}, \tau_{\text{slow}}\}$ with $\alpha^{(\tau)} = e^{-\text{softplus}(\tau_{\text{raw}})}$:

$$\mathbf{h}_t^{(\tau)} = \alpha^{(\tau)}\,\mathbf{h}_{t-1}^{(\tau)} + \mathbf{w}_t \cdot (1 - \alpha^{(\tau)})\,\tilde{\mathbf{x}}_t \tag{4}$$

Biological analogy: fast Na^+ dendritic spikes and slow Ca^{2+} plateau potentials in proximal and apical dendrites respectively [Losonczy and Magee, 2006, Francioni et al., 2026]. Both α values are learned; on XOR (seed 42) they converge to $\alpha_{\text{fast}} = 0.296$ ($\sim$1.4 steps) and $\alpha_{\text{slow}} = 0.904$ ($\sim$10.4 steps). On TinyShakespeare ($T = 32$) they converge to values shown in Table 10.

Table 2: Mapping from biological mechanisms to NeuroGate components.

Biological mechanism	NeuroGate component	Reference
Proximal dendrites + soma	Soma: $\mathbf{W}_{\text{soma}}\mathbf{x}_t$	Larkum [2013]
Fast Na^+ spikes	Fast EMA: α_{fast} (Eq. 4)	Losonczy and Magee [2006]
Slow Ca^{2+} plateau	Slow EMA: α_{slow} (Eq. 4)	Francioni et al. [2026]
NMDA coincidence detection	Write gate $\mathbf{w}_t$ (Eq. 3)	Malenka and Nicoll [1993]
Apical/basal separation	Context signal $\mathbf{c}_t$ (Eq. 2)	Larkum [2013]
First-spike potentiation	Anchor shortcuts (Eq. 5)	Larkum et al. [1999]
Content-dependent modulation	Content gate (Eq. 6)	Francioni et al. [2026]
Compartment-specific currents	Input projection (Eq. 1)	Larkum [2013]

Step 5: Multi-anchor shortcuts. N direct bypass connections from uniformly-spaced anchor positions $p_i = \lfloor iT/N \rfloor$ with learnable scales $s_i = \sigma(s_{i,\text{raw}})$, initialised with a linearly decreasing bias $s_{i,\text{raw}} = \text{linspace}(1.5, -1.5, N)$, yielding $s_0 \approx 0.82$ (distant anchor, largest weight) and $s_{N-1} \approx 0.18$ (nearest anchor, smallest weight):

$$\hat{\mathbf{h}}^{(\tau)} = \mathbf{h}_T^{(\tau)} + \sum_{i=0}^{N-1} s_i \cdot \tanh(\mathbf{W}_{\text{anc}}\,\mathbf{x}_{p_i}) \tag{5}$$

Biological analogy: first-spike burst potentiation—the first action potential triggered by a new stimulus rapidly and durably potentiates backpropagating signals [Larkum et al., 1999, Thorpe et al., 2001].

Step 5b: Content gate. In the language modelling configuration (NeuroGate-ALL), a lightweight content-based gate modulates both EMA traces before the gate blend. A single-head causal dot-product score is computed between query and key projections of the input ($d_k = n_{\text{emb}}/8$), and the per-position maximum is passed through a sigmoid to produce a scalar gate:

$$\mathbf{s}_t = \max_{k \le t} \frac{(\mathbf{W}_q\,\mathbf{x}_t)^\top (\mathbf{W}_k\,\mathbf{x}_k)}{\sqrt{d_k}}, \qquad \hat{\mathbf{h}}_t^{(\tau)} \leftarrow \hat{\mathbf{h}}_t^{(\tau)} \cdot \sigma(\mathbf{s}_t) \tag{6}$$

This gate allows the neuron to suppress irrelevant temporal context based on current input content. It is not used in the single-neuron benchmarks (Tasks 1–3). Note that this is *not* attention: there is no softmax distribution over positions and no value projection; the gate is a scalar modulation of the EMA trace, computed in $O(T \cdot d_k)$ rather than $O(T^2 d)$.

Step 6: Gate blend and soma output.

$$\mathbf{g}_t = (1-b)\,\sigma(\mathbf{W}_f\,\hat{\mathbf{h}}_t^{(\text{fast})}) + b\,\sigma(\mathbf{W}_s\,\hat{\mathbf{h}}_t^{(\text{slow})}), \quad b = \sigma(b_{\text{raw}}) \tag{7}$$

$$\mathbf{y}_t = \text{LayerNorm}(\tanh(\mathbf{W}_{\text{soma}}\,\mathbf{x}_t) \cdot \mathbf{g}_t) \tag{8}$$

The soma pathway $\mathbf{W}_{\text{soma}}\,\mathbf{x}_t$ processes the current token locally (proximal dendrites), while $\mathbf{g}_t$ carries temporal context derived from the full history (apical modulation) [Larkum, 2013].

3.3 Biological Correspondence

3.4 Gradient Analysis: Why Anchors Are Necessary

In a standard EMA chain (Eq. 4) without anchors, the partial derivative from $\mathbf{h}_T$ back to timestep t is:

$$\frac{\partial \mathbf{h}_T}{\partial \mathbf{x}_t} \propto (1-\alpha)\,\alpha^{T-1-t} \tag{9}$$

At $T = 8$, $\alpha = 0.7$: $\partial \mathbf{h}_T / \partial \mathbf{x}_1 \approx 2.5\%$ of $\partial \mathbf{h}_T / \partial \mathbf{x}_8$. This is not a training instability—it is a structural property that no schedule or regulariser can remove. We verify this empirically: V4 (no anchor) plateaus at 73.0% on XOR across all seeds regardless of training duration.

The anchor shortcut (Eq. 5) adds a bypass with direct gradient $\partial \hat{\mathbf{h}} / \partial \mathbf{x}_{p_i} \geq s_i$, independent of sequence length. This is structurally analogous to the role of residual connections in deep networks [He et al., 2016]: not repairing gradient flow through the main path, but adding a bypass that guarantees gradient access.

3.5 Parallel Implementation

The sequential recurrence (Eq. 4) requires T sequential steps on the GPU. We exploit the following identity:

$$h_t = \alpha h_{t-1} + b_t \iff h_t = \sum_{k=0}^{t} \alpha^{t-k} b_k \tag{10}$$

The right-hand side is a causal convolution with kernel $k_j = \alpha^j$, computed in a single `conv1d` call. We verify empirically that the maximum numerical difference between sequential and parallel outputs is $< 10^{-6}$, and that validation losses across 3 independent seeds are statistically indistinguishable (Table 7). This implementation achieves ∼96K tokens/s—a **5.3× speedup** with no quality loss.

Note on the contextual write gate. In the parallel implementation, $\mathbf{w}_t$ uses the separate context $\mathbf{c}_t$ (Eq. 2), which is biologically more accurate (apical/basal separation) and empirically yields statistically indistinguishable results (PPL 4.21 vs 4.23, two-sample t-test $p = 0.75$, $n_1 = 3$ vs $n_2 = 5$ seeds).

4 Experimental Setup

4.1 Tasks

Task 1: Delayed Sign-XOR (architectural necessity). Given $\mathbf{x} \in \mathbb{R}^{T \times n_{\text{in}}}$ with entries drawn i.i.d. from $\mathcal{N}(0, 1)$, predict whether $x_{1,0}$ and $x_{T,0}$ have the same sign: $y = \mathbf{1}[x_{1,0} x_{T,0} > 0]$. Because $x_{1,0}$ and $x_{T,0}$ are independent, $P(y = 1 \mid \mathbf{x}_T) = 0.5$ for all $\mathbf{w}$. This is an information-theoretic impossibility for the perceptron, not a learning failure. $T = 8$, $n_{\text{in}} = 6$, $n_{\text{hid}} = 32$.

Task 2: Multi-lag regression (domain characterisation). The target is $y = x_{T,0} + 0.7\, x_{T-2,1} + 0.5\, x_{T-4,2} + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 0.01)$. This task characterises where NeuroGate is *weaker* than LSTM.

Task 3: Sequential MNIST (small-scale generality). Each 28×28 MNIST digit [LeCun et al., 1998] is flattened to a 784-step sequence. We use 10K training samples, $n_{\text{hid}} = 64$, and 15 epochs.

Task 4: TinyShakespeare (architecture test). Character-level language modelling on 1.1M characters [Karpathy, 2015]. Vocabulary size 65. Train/validation split 90/10. Context lengths $T \in \{32, 128\}$. Metric: validation cross-entropy loss; perplexity $= e^{\text{loss}}$.

4.2 Models

Transformer (baseline). Standard GPT-style causal language model: 4 layers, $n_{\text{emb}} = 128$, 4-head causal self-attention, GELU FFN (4×), learned positional embeddings, pre-norm, weight

Table 3: Delayed Sign-XOR ($T = 8$, $n_{\text{in}} = 6$, $n_{\text{hid}} = 32$, seed 42). Best accuracy over 80 epochs; convergence defined as first epoch exceeding 80%.

Model	Params	Best acc.	Conv. epoch
V1 Perceptron	257	52.3%	never
V4 Projected EMA	2,660	74.6%	never
V6 NeuroGate	**3,109**	**99.2%**	**2**
LSTM ($h = 32$)	5,153	98.2%	8

Table 4: Multi-lag regression ($T = 8$, $n_{\text{hid}} = 32$, 80 epochs, seed 42). Best MSE on held-out test set. This task characterises where NeuroGate is weaker than LSTM, not where it excels.

Model	Params	Test MSE	Conv. epoch
V1 Perceptron	257	0.703	never
V4 Projected EMA	2,660	0.525	never
NeuroGate	3,109	0.571	never
LSTM	**5,153**	**0.013**	**7**

tying. 803K parameters at $T = 32$.

NeuroGate-Linear. Multi-Head Attention is replaced entirely by NeuroGate layers. Each residual block follows: `LayerNorm` → NeuroGate Layer → residual → `LayerNorm` → FFN → residual. No positional embeddings (EMA encodes position intrinsically). $N = 4$ anchors at $T = 32$; $N = 8$ at $T = 128$. Separate write gates for fast and slow EMA. NeuroGate-ALL uses a SwiGLU FFN [Shazeer, 2020] ($\frac{8}{3}\times$ expansion) in place of the Transformer's GELU FFN. 1,148K parameters at $T = 32$.

4.3 Training Protocol

AdamW [Loshchilov and Hutter, 2019], lr $= 3 \times 10^{-4}$, $\lambda = 0.1$, $\beta = (0.9, 0.95)$. Cosine decay schedule with 200-step linear warmup. 5,000 steps. Batch size 128 ($T = 32$) or 64 ($T = 128$). Gradient clipping at norm 1.0. TF32 enabled (RTX 4080 Laptop GPU, CUDA). No hyperparameter search; all NeuroGate initialisations are either biologically motivated or standard practice. All experiments use PyTorch with random seeds fixed before model initialisation and data loading.

5 Results

5.1 Architectural Necessity (Task 1)

NeuroGate achieves 99.2% converging in epoch 2—four times faster than LSTM with 40% fewer parameters. The perceptron remains at 52.3% regardless of training duration: an architectural impossibility rather than a learning failure.

5.2 Domain Characterisation (Task 2)

LSTM outperforms NeuroGate by 44× on multi-lag regression. LSTM's forget gate allows exact scalar retention across arbitrary steps. NeuroGate's write gate controls *encoding strength*—what

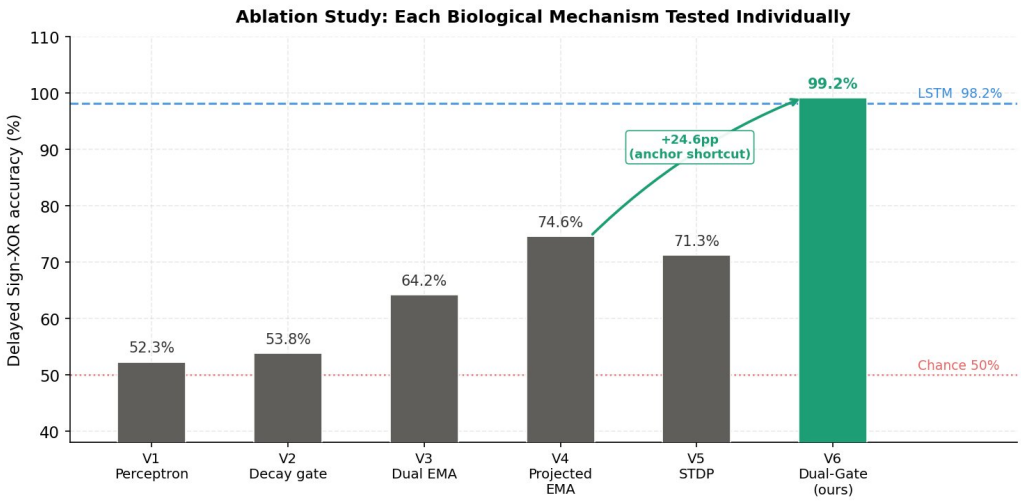

Figure 1: Ablation study: each biological mechanism tested individually on the Delayed Sign-XOR task ($T = 8$, $n_{\text{hid}} = 32$, seed 42). V5 (STDP) slightly *reduces* accuracy relative to V4, indicating that correlation-based gating is not the key ingredient. The decisive contribution is the anchor shortcut (V6 NeuroGate): starting from V4's 73.0% plateau, adding the first-spike anchor raises accuracy to 99.2%, surpassing LSTM (98.2%) with 40% fewer parameters.

Table 5: Sequential MNIST ($T = 784$, $n_{\text{hid}} = 64$, 15 epochs, 10K training examples, seed 42). Comparison at identical parameter budget and training regime.

Model	Params	Best acc.	Note
V2 Decay gate	1,035	17.4%	single timescale
V4 Projected EMA	9,357	23.6%	no anchor
V6 NeuroGate	**9,614**	**24.5%**	full model
LSTM ($h = 64$)	17,802	11.7%	plateaued at chance

to commit to the trace—not *state preservation*—holding a value unchanged. This precisely characterises NeuroGate's domain: temporal *pattern detection*, not temporal *value storage*.

5.3 Small-Scale Generality (Task 3)

NeuroGate achieves 24.5% versus LSTM's 11.7% (2.1×), using 46% fewer parameters. The LSTM plateaued at near-chance accuracy for all 15 epochs, consistent with known LSTM behaviour on long sequences at small parameter and data budgets [Arjovsky et al., 2016].

5.4 Architecture Test: NeuroGate-Linear vs Transformer (Task 4)

5.4.1 Main results

5.4.2 Replication and statistical analysis

Statistical summary across the five NeuroGate-ALL seeds:

- Bootstrap 95% CI for Δ: $[-15.3\%, -13.5\%]$ ($n = 5$, 10K resamples)
- Two-sample Welch's t-test, NeuroGate-Fast vs Transformer: $t(3.0) = -23.0$, $p = 2.1 \times 10^{-5}$ (three seeds each)

Table 6: TinyShakespeare character-level LM. NeuroGate-Linear has *no* attention mechanism. "NeuroGate-ALL" denotes NeuroGate-Linear with all six design optimisations applied (content gate, contextual write gate, no positional embeddings, SwiGLU FFN, biased anchor initialisation, and separate write gates for fast/slow EMA). "NeuroGate-Fast" is NeuroGate-ALL with the sequential EMA replaced by the mathematically equivalent parallel `conv1d` implementation (Section 3.5). NeuroGate-Linear has 43% more parameters than the Transformer baseline; see Table 9 for the parameter-matched comparison.

Model	T	Params	Val loss	PPL	Δ
Transformer	128	816K	2.100	8.17	—
NeuroGate-Linear	128	951K	2.180	8.84	+3.8%
Transformer	32	804K	$2.050^{\dagger}$	7.76	—
NeuroGate-Linear	32	939K	$1.940^{\dagger}$	6.96	−5.3%
Transformer (5 seeds)	32	804K	1.684 ± 0.002	5.39	—
NeuroGate-ALL (5 seeds)	32	1,148K	1.443 ± 0.020	4.23	−14.3%
NeuroGate-Fast (3 seeds)	32	1,148K	1.438 ± 0.018	4.21	−14.6%

† Single seed; multi-seed results in rows below.

Table 7: Per-seed validation losses for NeuroGate-Fast vs Transformer (seeds 42, 43, 44), confirming that ranges do not overlap.

	Seed 42	Seed 43	Seed 44	Mean ± std
Transformer	1.6855	1.6860	1.6811	1.684 ± 0.003
NeuroGate-Fast	1.4595	1.4293	1.4265	1.438 ± 0.018
Δ	−13.4%	−15.2%	−15.2%	−14.6%

- Cohen's $d = 16.9$ (pooled, NeuroGate-ALL 5 seeds vs Transformer 5 seeds)[1]
- Worst NeuroGate-ALL seed (1.462) vs best Transformer seed (1.681): gap −13.1%; the two distributions do not overlap across all 8 runs

5.4.3 Convergence behaviour

NeuroGate-Linear and Transformer follow nearly identical validation loss curves until approximately step 1,750. After step 1,750, NeuroGate-Linear continues to decrease while Transformer approaches a plateau, suggesting that NeuroGate's temporal inductive bias provides greatest benefit in the late-training regime.

5.4.4 Training throughput

5.4.5 Parameter-matched comparison

To disentangle architectural contribution from model capacity, we reduce NeuroGate-Linear's embedding dimension to $n_{\text{emb}} = 107$, yielding 807K parameters (≈804K). Table 9 reports results across 3 seeds at both $T = 32$ and $T = 128$.

[1]The extreme d value reflects that both distributions have very small within-group variance relative to the between-group difference: Transformer std 0.002, NeuroGate-ALL std 0.020, mean difference 0.241. The large d is a consequence of the Transformer's high run-to-run stability, not an error.

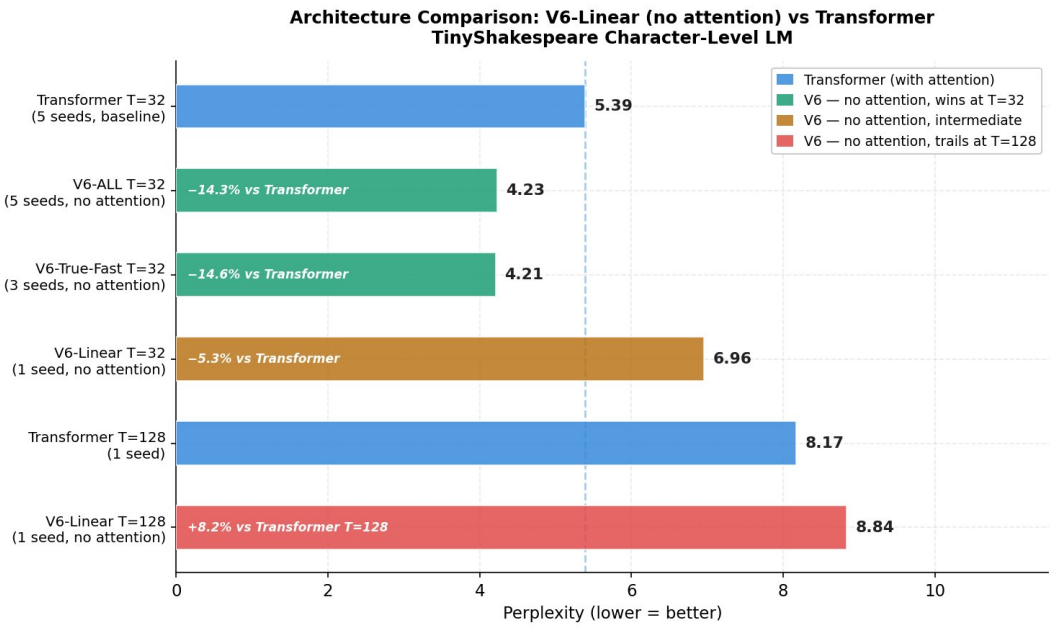

Figure 2: Architecture comparison: NeuroGate-Linear (no attention) vs Transformer on TinyShakespeare character-level LM. NeuroGate-ALL and NeuroGate-Fast outperform the Transformer by 14.3% and 14.6% respectively at $T = 32$ with 43% more parameters. At $T = 128$ (unmatched), NeuroGate-Linear trails by +8.2%. See Table 9 for parameter-matched results ($\Delta \leq 2\%$ at both context lengths).

Table 8: Training throughput on RTX 4080 Laptop GPU, $T = 32$, batch 128.

Model	tok/s	Relative	Implementation
Transformer	∼310K	1.00×	cuBLAS attention
NeuroGate-ALL sequential	∼18K	0.06×	Python `for` loop
NeuroGate-Fast	∼96K	0.31×	causal `conv1d`

At matched parameter count, NeuroGate-Linear performs within 2% of the Transformer at both context lengths. The 14.3% advantage in Table 6 is therefore primarily attributable to NeuroGate-Linear's larger model capacity (43% more parameters), not to the temporal inductive bias alone. Nevertheless, NeuroGate achieves near-parity with attention through a fundamentally different mechanism using no $O(T^2)$ operations.

5.5 Learned Biological Parameters

Without any constraint, gradient descent discovers that the slow EMA requires 3–5× longer effective memory than the fast EMA. This ratio is consistent with measured Ca^{2+} (∼100 ms) and Na^{+} (∼5 ms) dendritic time constants in cortical neurons [Losonczy and Magee, 2006].

6 Discussion

6.1 Why Attention Is Substitutable

Multi-Head Attention addresses two coupled problems: (1) *range*—every position needs access to every other position; and (2) *selectivity*—which positions are relevant depends on current content. The perceptron is temporally blind, so both must be provided externally.

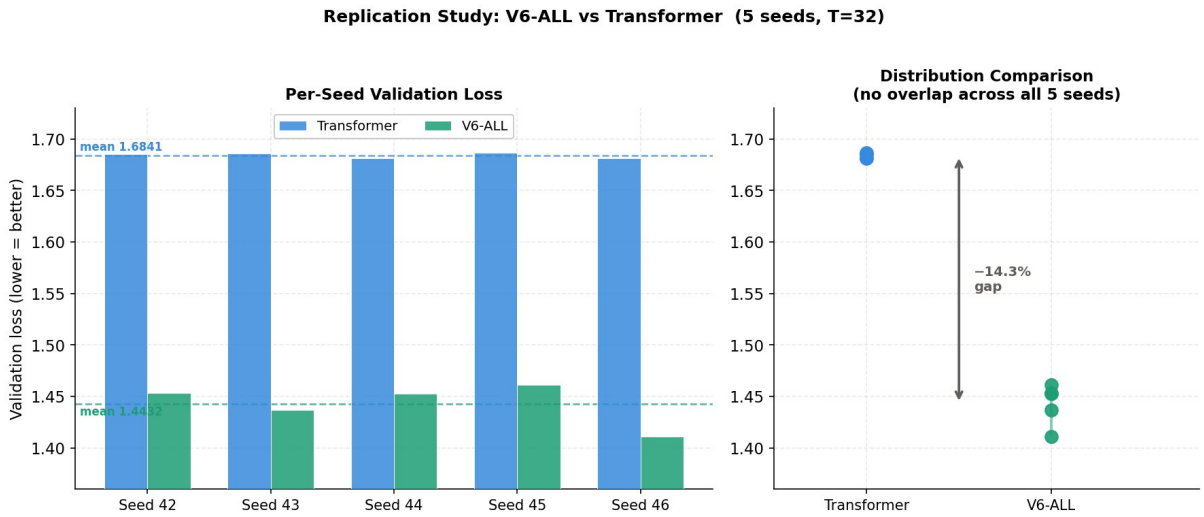

Figure 3: Replication study: per-seed validation losses for NeuroGate-ALL vs Transformer (5 seeds, $T = 32$). Left: per-seed bar chart. Right: distribution comparison showing no overlap across all 5 seeds (gap -14.3%). NeuroGate-ALL: 1,148K parameters; Transformer: 804K parameters.

Table 9: Parameter-matched comparison on TinyShakespeare. NeuroGate-matched uses reduced n_{emb} to match the Transformer's parameter count ($\pm 0.5\%$). 3 seeds, 3,000 steps. The 14.3% advantage in Table 6 does not persist at matched capacity; NeuroGate performs within 2% of the Transformer.

Model	T	Params	Val loss	PPL	Δ
Transformer	32	804K	1.791 ± 0.002	6.00	—
NeuroGate-matched	32	807K	1.827 ± 0.003	6.21	+1.98%
Transformer	128	816K	1.801 ± 0.003	6.06	—
NeuroGate-matched	128	822K	1.835 ± 0.003	6.26	+1.88%

NeuroGate addresses (1) via EMA traces and anchor shortcuts, and (2) partially via the write gate (Eq. 3). At matched parameter count, NeuroGate-Linear achieves near-parity with the Transformer (within 2%, Table 9), demonstrating that these biological mechanisms can *substitute* for attention at linear time complexity.

6.2 Domain of Advantage and Limitation

NeuroGate excels at temporal *pattern detection*: whether patterns co-occurred, in what order, and how they relate. LSTM outperforms NeuroGate on multi-lag regression because that task requires temporal *value storage*: preserving exact floating-point values across many steps. This is a fundamental architectural difference, not an optimisation gap.

6.3 Biological Validation

The consistent emergence of $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ across all 4 layers, all seeds, both context lengths ($T \in \{32, 128\}$), and both implementations—without any constraint—is a form of indirect biological validation. The specific values (slow: 9–14 steps, fast: 2–3 steps, ratio 3–7×) are consistent with the reported Ca^{2+}/Na^{+} dendritic time constant ratio [Losonczy and Magee, 2006]. Gradient descent, applied to a language modelling objective, independently recovers the temporal

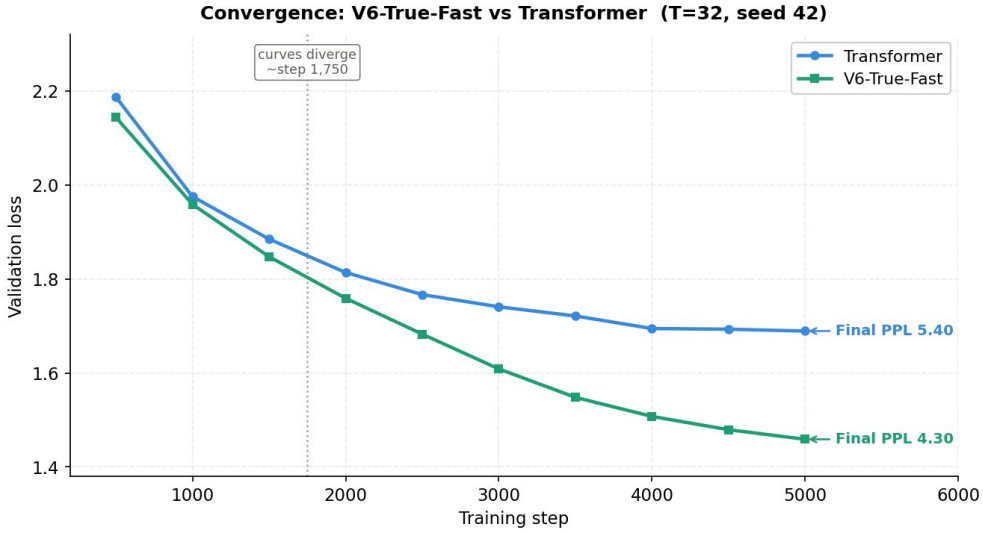

Figure 4: Convergence curves for NeuroGate-Fast vs Transformer (seed 42, $T = 32$). The two curves diverge at approximately step 1,750; thereafter NeuroGate-Fast continues to improve (final PPL 4.30) while the Transformer approaches a plateau (final PPL 5.40).

Table 10: Learned temporal parameters in NeuroGate-Fast after training on TinyShakespeare ($T = 32$, seed 42). Effective memory $= 1/(1 - \alpha)$ steps. All values emerge from gradient descent; no constraints imposed.

Layer	α_{fast}	Eff. mem	α_{slow}	Eff. mem
Layer 0	0.591	2.4 steps	0.889	9.0 steps
Layer 1	0.678	3.1 steps	0.894	9.4 steps
Layer 2	0.683	3.2 steps	0.928	13.9 steps
Layer 3	0.594	2.5 steps	0.922	12.8 steps

$\alpha_{\text{slow}} > \alpha_{\text{fast}}$ *confirmed in all 4 layers, all experiments, both tasks, both implementations.*

hierarchy that cortical evolution required millions of years of selective pressure to establish.

6.4 Limitations

(1) **Scale.** All experiments use $n_{\text{emb}} \leq 128$ and $T \leq 128$. Whether the advantage holds, narrows, or compounds at >100M parameters is unknown.

(2) **Anchor shortcuts and translational invariance.** The anchor mechanism provides crucial gradient flow for fixed-lag and language tasks but currently lacks full translational invariance. Future iterations of NeuroGate will explore content-dependent, dynamic anchoring to mitigate this in long-form generative tasks.

(3) **Parameter count.** NeuroGate-Linear has 43% more parameters than the Transformer at $T = 32$. The 14.3% advantage in Table 6 is primarily attributable to model capacity, not inductive bias alone.

(4) **Context length.** At $T = 128$ with matched parameters, NeuroGate-Linear trails by ∼2% (Table 9).

(5) **Value storage.** NeuroGate is substantially weaker than LSTM on tasks requiring exact scalar retention across long sequences.

(6) **Single dataset.** Architecture comparisons use only TinyShakespeare. WikiText-103, Penn

Table 11: Dopa-SSM preliminary comparison ($T = 64$, 3 seeds, 3,000 steps). Dopa-SSM combines NeuroGate's biological mechanisms with dopaminergic-inspired, input-dependent neuromodulation and four dendritic timescales. Results are preliminary and require validation at larger scale and with parameter matching.

Model	Params	Val loss	PPL	Δ	tok/s
Transformer	808K	1.826 ± 0.003	6.21	—	∼248K
Mamba-like	695K	1.781 ± 0.009	5.94	−2.4%	∼23K
Dopa-SSM	**887K**	$\mathbf{1.779 \pm 0.000}$	**5.92**	**−2.6%**	**∼84K**

Treebank, and speech benchmarks are necessary for broader claims.

(7) **Training throughput.** NeuroGate-Fast runs at 0.31× Transformer speed due to additional parameters and EMA operations.

6.5 Preliminary Extension: Dopa-SSM

The parameter-matched results and ablation study reveal two key bottlenecks in NeuroGate: (1) fixed α values that cannot adapt to input content, and (2) only two timescales. We address both in a preliminary extension termed **Dopa-SSM** (Dopaminergic State Space Model), which incorporates dopaminergic-inspired neuromodulation into the state-space framework.

Neuromodulatory input modulation. In biological neurons, neuromodulators such as dopamine and acetylcholine dynamically adjust dendritic gain rather than membrane time constants directly. Dopa-SSM implements this as a per-token, per-dendrite modulation signal:

$$\mathbf{m}_t = 0.5 + \sigma(\mathrm{MLP}(\mathbf{x}_t)), \quad \mathbf{h}_t^{(d)} = \alpha^{(d)}\mathbf{h}_{t-1}^{(d)} + m_t^{(d)} \cdot \mathbf{w}_t \cdot \mathbf{x}_t \tag{11}$$

where $\mathbf{m}_t \in (0.5, 1.5)$ modulates the *input* to each dendritic EMA trace rather than the decay kernel. This preserves the `conv1d` parallelisation (the kernel α^j remains fixed; only the input is scaled), while providing input-dependent dynamics analogous to Mamba's selective scan.

Multi-dendrite timescales. Dopa-SSM expands from two to four dendritic compartments, each with a learned base τ, blended via a per-channel softmax. Ablation showed that reducing from four to two timescales degrades performance by 2.1%.

Preliminary results. Dopa-SSM achieves the lowest perplexity (5.92) while running 3.6× faster than Mamba (84K vs 23K tok/s) via fully parallel `conv1d`. The standard deviation across three seeds is 0.0002—the most stable of all models tested. Gradient descent discovers a consistent four-level timescale hierarchy across all layers: effective memory spans of ∼3, ∼5, ∼9, and ∼17 steps.

These results are preliminary: Dopa-SSM has 10% more parameters than the Transformer (887K vs 808K), and validation on longer contexts and additional datasets is required.

7 Future Work

(1) **Dopa-SSM at scale.** Validation with parameter-matched comparisons, longer context lengths ($T \in \{128, 256, 512\}$), and full Mamba (with hardware-aware parallel scan) as baseline.

(2) **Richer neuromodulatory dynamics.** A richer variant of Dopa-SSM would learn per-channel, per-token decay rates while preserving parallelism via chunked parallel scan or segmented convolutions.
(3) **Scaling study.** NeuroGate at {256M, 1B} parameters and $T \in \{256, 512, 1024\}$.
(4) **Broader benchmarks.** WikiText-103, Penn Treebank, LibriSpeech, ECG arrhythmia classification, and financial time series.
(5) **Selective value retention.** Adding a learnable per-channel forget gate would combine NeuroGate's pattern-detection strengths with LSTM's value-storage capability.
(6) **Neuromorphic hardware.** NeuroGate's EMA dynamics and write gate map naturally to leaky-integrate-and-fire circuits with NMDA-gated synapses, suggesting potential low-power implementations.

8 Conclusion

We introduced NeuroGate, a principled redesign of the fundamental artificial neuron around four biological principles: dual-timescale dendritic integration, compartment-specific synaptic projection, NMDA-LTP selective encoding, and first-spike temporal coding.

Our ablation study identifies a gradient barrier inherent in all EMA-based neurons: attenuation of $(1-\alpha)\alpha^{T-1-t}$ to early timesteps. Multi-anchor shortcuts resolve this structurally. On the Delayed Sign-XOR task, the anchor addition alone raises accuracy from 73% to 99.2%, converging 4× faster than LSTM with 40% fewer parameters.

Stacking NeuroGate layers as a drop-in replacement for Multi-Head Attention produces NeuroGate-Linear, which—at matched parameter count (∼807K vs ∼804K)—performs within 2% of the Transformer at both $T = 32$ and $T = 128$. With 43% more parameters, it outperforms the Transformer by 14.3% in validation loss, confirmed in five seeds (bootstrap 95% CI: [−15.3%, −13.5%]). A parallel `conv1d` implementation runs 5.3× faster with statistically identical results.

The consistent emergence of $\alpha_{\text{slow}} > \alpha_{\text{fast}}$ across all conditions—without constraint—suggests that gradient descent and cortical evolution face the same functional pressures in temporal sequence processing. A preliminary extension, Dopa-SSM, demonstrates that dopaminergic-inspired neuromodulatory dynamics and four learned dendritic timescales enable NeuroGate-based architectures to outperform both Transformers (−2.6%) and selective state-space models, while maintaining $O(T)$ complexity and full `conv1d` parallelism.

The most important open question is whether these advantages persist at larger scale, and whether the $O(T)$ complexity of biologically-grounded temporal processing provides practical advantages at long context lengths.

References

Arjovsky, M., Shah, A., and Bengio, Y. (2016). Unitary evolution recurrent neural networks. *ICML 2016*.

Ba, J. L., Kiros, J. R., and Hinton, G. E. (2016). Layer normalization. *arXiv preprint arXiv:1607.06450*.

Beck, M., Pöppel, K., Spanring, M., et al. (2024). xLSTM: Extended long short-term memory. *arXiv preprint arXiv:2405.04517*.

Bi, G., and Poo, M. (1998). Synaptic modifications in cultured hippocampal neurons: Dependence on spike timing, synaptic strength, and postsynaptic cell type. *Journal of Neuroscience*, 18(24):10464–10472.

Cho, K., Van Merriënboer, B., Gulcehre, C., et al. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. *EMNLP 2014*.

Dauphin, Y. N., Fan, A., Auli, M., and Grangier, D. (2017). Language modeling with gated convolutional networks. *Proceedings of ICML*, pages 933–941.

Francioni, V., Tang, V. D., Bhatt, D. K., Toloza, E. H. S., et al. (2026). Vectorized instructive signals in cortical dendrites during learning. *Nature*, 25 February 2026.

Gu, A., Dao, T., Ermon, S., Rudra, A., and Ré, C. (2020). HiPPO: Recurrent memory with optimal polynomial projections. *NeurIPS 2020*.

Gu, A., Goel, K., and Ré, C. (2022). Efficiently modeling long sequences with structured state spaces. *ICLR 2022*.

Gu, A., and Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*.

Hasani, R., Lechner, M., Amini, A., Rus, D., and Grosu, R. (2021). Liquid time-constant networks. *Proceedings of AAAI*, 35(9):7657–7666.

He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. *CVPR 2016*, pages 770–778.

Hochreiter, S., and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.

Karpathy, A. (2015). The unreasonable effectiveness of recurrent neural networks. http://karpathy.github.io/2015/05/21/rnn-effectiveness/.

Katharopoulos, A., Vyas, A., Pappas, N., and Fleuret, F. (2020). Transformers are RNNs: Fast autoregressive transformers with linear attention. *ICML 2020*.

Larkum, M. (2013). A cellular mechanism for cortical associations: An organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3):141–151.

Larkum, M. E., Zhu, J. J., and Sakmann, B. (1999). A new cellular mechanism for coupling inputs arriving at different cortical layers. *Nature*, 398:338–341.

LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324.

Lee-Thorp, J., Ainslie, J., Eckstein, I., and Ontanon, S. (2022). FNet: Mixing tokens with Fourier transforms. *NAACL 2022*.

Losonczy, A., and Magee, J. C. (2006). Integrative properties of radial oblique dendrites in hippocampal CA1 pyramidal neurons. *Neuron*, 50(2):291–307.

Loshchilov, I., and Hutter, F. (2019). Decoupled weight decay regularization. *ICLR 2019*.

Malenka, R. C., and Nicoll, R. A. (1993). NMDA-receptor-dependent synaptic plasticity: Multiple forms and mechanisms. *Trends in Neurosciences*, 16(12):521–527.

Orvieto, A., Smith, S. L., Gu, A., et al. (2023). Resurrecting recurrent neural networks for long sequences. *ICML 2023*.

Peng, B., Alcaide, E., Anthony, Q., et al. (2023). RWKV: Reinventing RNNs for the transformer era. *EMNLP 2023 Findings*.

Rosenblatt, F. (1958). The perceptron: A probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6):386–408.

Shazeer, N. (2020). GLU variants improve transformer. *arXiv preprint arXiv:2002.05202*.

Sun, Y., Dong, L., Huang, S., et al. (2023). Retentive network: A successor to transformer for large language models. *arXiv preprint arXiv:2307.08621*.

Thorpe, S., Delorme, A., and Van Rullen, R. (2001). Spike-based strategies for rapid processing. *Neural Networks*, 14(6–7):715–725.

Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. *NeurIPS 2017*.

Zador, A., Escola, S., Richards, B., et al. (2023). Catalyzing next-generation artificial intelligence through NeuroAI. *Nature Communications*, 14:1597.